

2.8inch Smart Display

WT32S3-28S PRO (ZX2D80CE02S)



Changelog:

date	Change of person	illustrate
2025/06/11	hades	Create a document

Features:

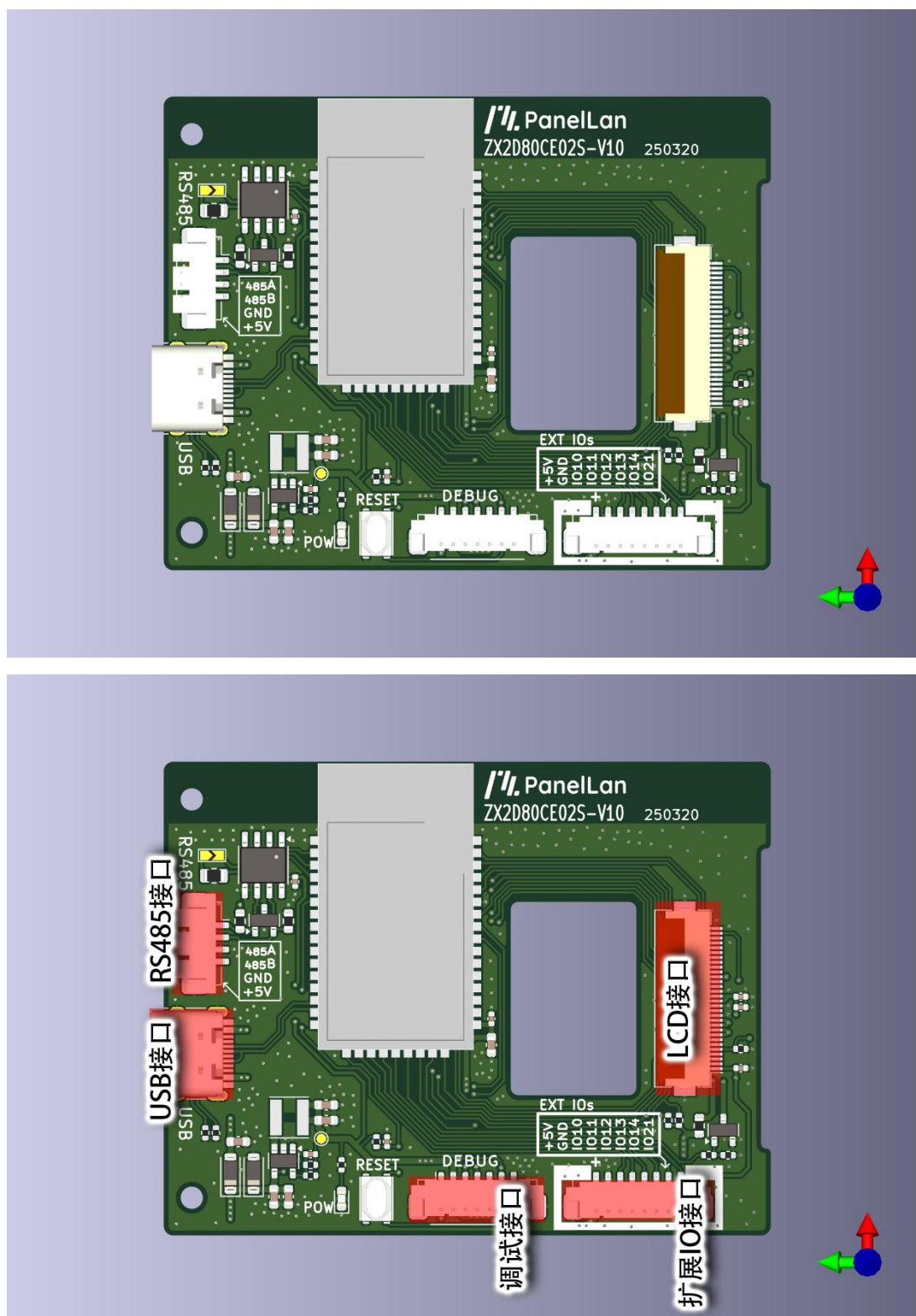
1. Rapid development

Core Item List (Tab.0):

serial number	name	Model	remark
1	ESP32-S3	WT32-S3-WROVER-N8R2	

Hardware interface:

Hardware interface diagram:



Hardware Interface (Figure 1 Hardware Interface Diagram).

Interface description:

[1] Debug Interface (Tab.1).

Pin	description	Module PIN	Voltage range	remark
1	+5V	-	5V	
2	+3.3V	-	3.3V	For reference use, not for power input
3	ESP_TXD	TXD0	3.3V TTL	
4	ESP_RXD	RXD0	3.3V TTL	
5	EN	EN	0-3.3V	Chip enabled
6	BOOT	GPIO 0	0-3.3V	
7	GND	-	0V	earthing

[2] Extended IO ports (Tab.2).

Pin	description	Module Pin	Voltage range	remark
1	+5V	-	5V±5%	Power supply or output power supply
2	GND	-	0V	earthing
3	EXT_IO1	GPIO 10	0-3.3V	Extended IO
4	EXT_IO2	GPIO 11	0-3.3V	
5	EXT_IO3	GPIO 12	0-3.3V	
6	EXT_IO4	GPIO 13	0-3.3V	
7	EXT_IO5	GPIO 14	0-3.3V	
8	EXT_IO6	GPIO 21	0-3.3V	

[3] LCD interface (Tab.3).

description	Module Pin	remark
BL_PWM	GPIO 47	Backlight control, high effectiveness
LCD_RESET	GPIO 3	LCD reset, multiplexed with touch reset
LCD_RS	GPIO 18	Command/data selection
LCD_WR	GPIO 17	Write the clock
LCD_TE	GPIO 38	Frame synchronization
LCD_DB0	GPIO 16	LCD data interface, 8-bit mcu(8080)
LCD_DB1	GPIO 40	
LCD_DB2	GPIO 15	
LCD_DB3	GPIO 7	
LCD_DB4	GPIO 41	
LCD_DB5	GPIO 42	
LCD_DB6	GPIO 2	
LCD_DB7	GPIO 1	
TP_INT	GPIO 48	Touch Interrupt
TP_SDA	GPIO 8	Touch IIC data
TP_SCL	GPIO 9	Touch the IIC clock
TP_RST	GPIO 3	Touch reset, multiplexed with LCD reset

[4] RS485 connector (Tab.4).

Pin	description	remark
1	RS485-A	RS485 bus
2	RS485-B	
3	GND	earthing
4	+5V	Power supply or output power supply

Interface Package :

Description of the interface	Interface encapsulation	remark
Debug interfaces	Molex SD-53398-0771 (7Pin) WAFER MX1.25 7P Stand-Up Sticker	MX1.25
Expand the I/O interface	Molex SD-53398-0871 (8Pin) WAFER MX1.25 8P Stand-up stickers	MX1.25
Speaker interface	Molex SD-53398-0271 (2Pin) WAFER MX1.25 2P Stand-Up Sticker	MX1.25
RS485 interface	Molex SD-53261-0471 (4Pin) WAFER MX1.25 4P Lying Sticker	MX1.25

Hardware Peripherals :

The name of the peripheral	description
LCD with touch	
RS485	Serial port with flow control IO

RS485 :

description	Module PIN
RXD	GPIO 4
RTS	GPIO 5
TXD	GPIO 6

Schematic:

PDF files are available in the attached list

Specifications :

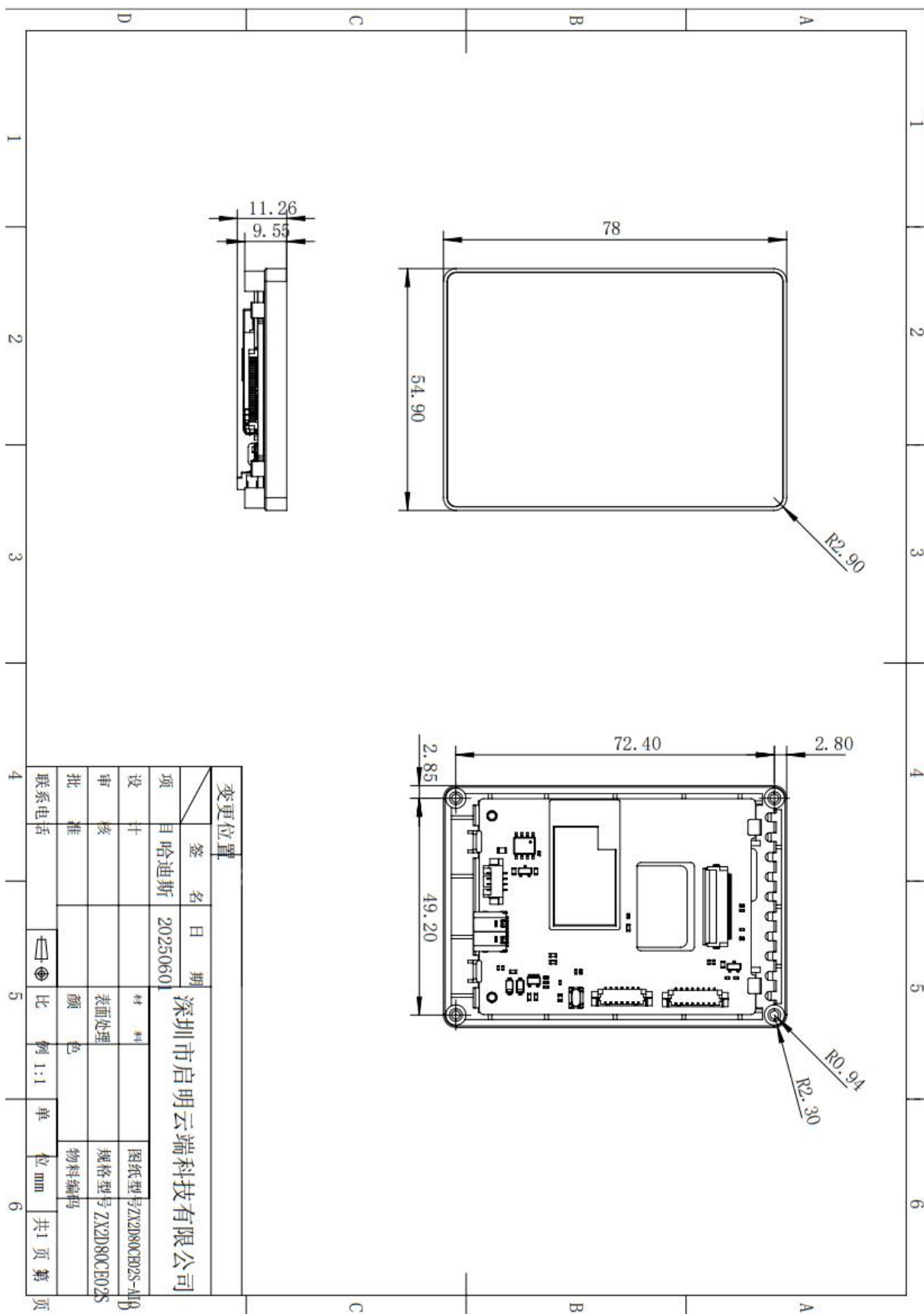
[1] Display Parameters (Tab.1)

Display type	Normally black
Driver IC model	ST7789
Visual angle	ALL VIEWING
Resolution	240*320
Interface	MCU 8 bit
Color	262K
Backlight mode	Transmissive mode

[2] Electrical Parameters (Tab.2).

parameter	Test conditions	mini mum	Typical	maximum	unit
Operating voltage	-	4.5	5.0	5.5	V
Operating current	USB provides 5V with maximum backlight brightness		150		mA

Exterior structure diagram (Fig.2).



Firmware burning:

1. Connect the downloader (ZXACC-ESPDB) via USB-Type C and connect the board to the downloader (ZXACC-ESPDB) with an MX1.25-7P cable. Since the downloader (ZXACC-ESPDB) does automatic data flow control, the firmware can be automatically downloaded through the ESP32 Flash Download Tools.
2. As shown in Figure 4 (Fig.4) on the right: select the firmware path to be burned at 1, the address is usually 0X00, and remember to tick the front after setting; Select 40MHz for the system clock at 2; Select 3 locations for the size of Flash as 32Mbit; Select SPI MODE to DIO mode at 4; Select the port number recognized by the current board on the computer at 5; Select the serial port baud rate at 6 (the higher the value, the faster the firmware download rate, the maximum support is 1152000bps);
3. After completing the previous configuration, click at 7 to start burning the firmware.
4. After completing the above two steps, press the reset button on the back of the board to start running the firmware you just flashed.

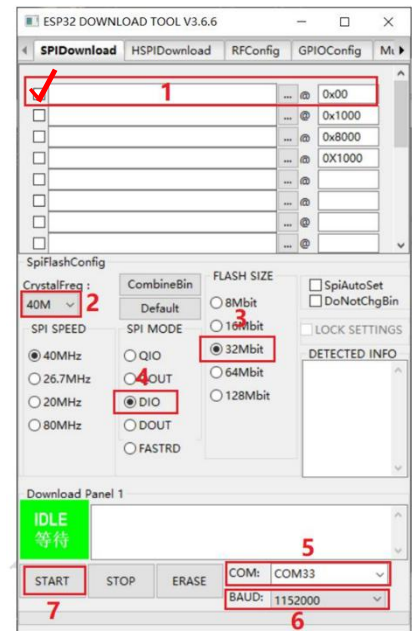


Figure 4 (Fig.4)

Software reference:

URL: <http://doc.panel-tag.cn/ESP32-S3/flash.html>

Online Platform User Manual: <http://doc.panel-tag.cn/ESP32-S3/index.html>

LVGL official website link: <https://lvgl.io/>

LVGL is recommended to be used in version v8.3.1. The recommended UI development tool is NXP-GUI Guider

GUI-Guider

<https://www.nxp.com/design/design-center/software/development-software/gui-guider:GUI-GUIDER>